Photovoltaic DC Arc-Fault and Connector Fire - Safety Bulletin

1. Why PV DC arcs are dangerous [p1]

Alternating current crosses zero 100-120 times per second, which self-extinguishes most arcs. Photovoltaic DC has NO current zero-crossing, so once a series arc strikes at a degraded connector or joint it is self-sustaining. PV arcs reach several thousand degrees C and readily ignite cable insulation, the connector body and the surrounding structure. Connector and joint overheating is among the leading causes of PV system fires.

2. Arc-fault types

Type	Where	Trigger
Series arc	in-line joint / connector	high-resistance contact under load
Parallel arc	between conductors	insulation breakdown
Ground arc	conductor to frame / earth	insulation or earth fault

3. The failure chain

poor crimp / cross-mating / water ingress -> high contact resistance -> I2R heating -> housing melt and contact oxidation -> intermittent contact -> series arc -> fire. The melted MC4 connector a field engineer photographs is the visible mid-point of this chain.

4. Recognising it

Channel	Sign
Visual	melted / charred connector, discoloured insulation, soot, frayed copper
Thermal	connector temperature high vs neighbours on an IR scan
Electrical	AFCI trip, string-current imbalance, intermittent drop-outs
Olfactory	burning or ozone smell

5. Isolation before any work

1. Treat all strings as LIVE in any daylight.
2. Open the DC isolator / switch-disconnector under NO LOAD - do not break the circuit by pulling a loaded connector.
3. Apply lockout / tagout.
4. Confirm de-energised with a meter rated for the system DC voltage (up to 1500 V).
5. Never separate or cut a connector under load - that itself draws an arc.

6. Code requirements

- NEC 690.11 - DC arc-fault circuit protection (AFCI) is required on PV systems; treat AFCI events as genuine arc faults.
- NEC 690.12 - rapid shutdown of module-level and array conductors.
- Local electrical and fire codes may impose additional isolation and signage requirements.

7. PPE and tools

- Arc-rated clothing and face protection.
- Gloves and insulated tools rated above the array open-circuit voltage (Voc).
- No work on a suspected arc fault in wet conditions.

8. Remediation

1. Isolate and make safe (section 5).

2. Cut out the damaged connector PAIR.
3. Re-terminate with matched, genuine connectors to the install spec.
4. IR-scan the adjacent connectors and the combiner box.
5. Root-cause it (poor crimp vs cross-mating vs ingress) and audit the array for the same defect.

9. Do / Don't

- DO isolate under no-load; DO replace BOTH halves; DO thermal-scan the neighbours.
- DON'T break a loaded DC connection; DON'T re-use a heat-damaged connector; DON'T mix connector brands.